

AHMED GAMAL ARFA

Data Analyst | Data Scientist | Data Engineer | AI Engineer

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PROFESSIONAL SUMMARY

Data and AI professional with hands-on experience across data analysis, business intelligence, data science, data engineering, and AI engineering. Microsoft Certified Power BI Data Analyst (PL-300) and Fabric Analytics Engineer Associate, with applied experience building executive Power BI dashboards, SQL data-quality and ETL pipelines, machine learning classification models, and AI-powered production features including OCR-based document data extraction and LLM automation. Built 4+ executive dashboards reducing reporting preparation time by approximately 60%, engineered SQL pipelines processing 4,000+ records, and developed classification models reaching 97% accuracy. Strong at translating business requirements into data models, KPIs, actionable insights, and intelligent automation within production-oriented systems.

CORE SKILLS

Data Analysis & BI: Power BI, DAX, Power Query, Microsoft Fabric, KPI Design, Dashboard Development, Data Visualization, Data Storytelling, Advanced Excel, Pivot Tables, Reporting

Data Science: Python, Pandas, NumPy, R, Scikit-Learn, Classification, Clustering, Regression, Exploratory Data Analysis, Statistical Analysis, Feature Engineering

Data Engineering: SQL, SQL Server, T-SQL, PostgreSQL, MySQL, ETL Pipelines, Data Modeling, Data Warehousing, Star Schema, Data Cleaning, Data Validation

AI Engineering: PyTorch, NLP, Computer Vision, OCR, Retrieval-Augmented Generation (RAG), LLM Integration, Model Training & Evaluation, AWS AI/ML Fundamentals

Backend & APIs: FastAPI, REST APIs, JWT Authentication & Authorization, React Integration, File Handling, Database Operations

Tools & Collaboration: Git, GitHub, VS Code, Jupyter Notebook, Linux, Stakeholder Communication, Requirements Gathering, Documentation, Teamwork, Mentoring

CERTIFICATIONS

- Microsoft Certified: Fabric Analytics Engineer Associate | Microsoft | Earned July 14, 2026 | Expires July 15, 2027
- Microsoft Certified: Power BI Data Analyst Associate (PL-300) | Microsoft | Earned July 4, 2026 | Expires July 5, 2027
- AWS: Becoming a Machine Learning Engineer — Learning Path Certificate of Completion | Manara | August 23, 2026
- AWS: AI Practitioner — Learning Path Certificate of Completion | Manara | August 26, 2026
- Google Data Analytics Professional Certificate | Google | 2026

All 8 courses completed: Foundations: Data, Data, Everywhere; Ask Questions to Make Data-Driven Decisions; Prepare Data for Exploration; Process Data from Dirty to Clean; Analyze Data to Answer Questions; Share Data Through the Art of Visualization; Data Analysis with R Programming; Capstone: Complete a Case Study.

- Supervised Machine Learning: Regression and Classification | Stanford University & DeepLearning.AI | 2026
- Introduction to Retrieval Augmented Generation (RAG) | Duke University | 2026
- Introduction to Image Processing and Image Segmentation | MathWorks | 2026
- Introduction to Python, Introduction to SQL, and Intermediate SQL | DataCamp | 2025
- Database Fundamentals | Information Technology Institute (ITI) | 2025

PROFESSIONAL EXPERIENCE

AI & Data Analysis Intern — Validation Team

Ministry of Communications and Information Technology (MCIT), Cairo, Egypt

September 6, 2026 – Present

- Contribute to production-oriented validation systems, working across backend services, frontend integration, and AI-powered features.
- Develop backend services and REST APIs using FastAPI, including file handling and database operations.
- Implement JWT-based authentication and authorization to secure application endpoints.
- Build OCR pipelines for ID card detection and extraction of structured data for downstream validation and processing.
- Integrate React frontend components with backend APIs to deliver end-to-end application functionality.
- Apply data validation rules and database operations to ensure data accuracy and integrity across system workflows.

Data Analyst and AI Trainee

Ministry of Communications and Information Technology (MCIT) – Digilians Program, Cairo, Egypt 2025 – Present

- Built 4+ executive Power BI dashboards across Sales, Healthcare, and HR using DAX measures, reducing reporting preparation time by approximately 60%.
- Engineered SQL data-cleaning pipelines processing 4,000+ records with deduplication, validation rules, and audit checks to improve data reliability.
- Translated stakeholder requirements into KPI trees and drill-through reports to support faster, evidence-based business decisions.
- Performed data cleaning, transformation, analysis, and visualization using Power BI, Excel, SQL, and Python.
- Developed analytical reports and data models for cross-functional business scenarios in a deadline-driven environment.

Data and Technology Mentor (Volunteer)

MCIT Digilians Community, Cairo, Egypt

2024 – Present

- Mentored 15+ junior students in SQL, Python, and Power BI fundamentals through weekly technical sessions.
- Reviewed dashboards and analytics projects, providing structured feedback on data modeling, KPI design, and visual storytelling.
- Organized knowledge-sharing sessions covering AI and machine learning workflows.

SELECTED PROJECTS

NILEX.AI — Smart Green Ecosystem for Egyptian Agricultural Exports | Graduation Project (Team of 4)

Python | YOLOv8n | ResNet-50 | PyTorch | FastAPI | PostgreSQL | Microsoft Azure | Power BI | RAG | Flutter

- Built an AI decision-support platform for citrus export quality grading, combining YOLOv8n fruit detection ($mAP@0.50:0.95 = 0.81$) with a ResNet-50 disease classifier ($Macro-F1 = 0.978$) at under 3-second inference latency.
- Deployed a microservices architecture on Microsoft Azure with PostgreSQL, FastAPI ingestion, and Power BI KPI dashboards.
- Extended the platform with a RAG agricultural knowledge assistant, Arabic speech-to-text voice interface, and a multi-agent system for export, finance, and carbon-footprint monitoring.
- Live deployment: nilex-ai-production.eastus.cloudapp.azure.com

Hospital Patient Flow Command Center | Power BI | SQL

- Executive dashboard tracking 300 patients across 8 departments and EGP 37.44M in revenue, reducing executive review time by approximately 50%.

Superstore Sales Dashboard | Power BI | DAX

- Interactive sales report covering 4,117+ orders with cross-filtering, geographic drill-down, and profitability KPIs.

SQL Data Quality Pipeline | SQL Server | T-SQL — Deduplication, validation, semantic layers, and audit checks applied to 4,000+ records.

Machine Learning Classification Pipeline | Python | Scikit-Learn — End-to-end preprocessing, feature engineering, and explainable evaluation, reaching 97% accuracy.

AI Insight Automation Engine | Python | Pandas | LLM — Profiles datasets, detects anomalies, and generates narrative insights for stakeholders.

Customer Segmentation Lab | Python | K-Means — Unsupervised segmentation producing actionable behavioral customer groups.

EDUCATION

Professional Diploma in Data Analysis and Artificial Intelligence

Egyptian Military Academy in partnership with MCIT Digilians

September 2025 – August 2026 (In Progress)

- Coursework: Python for Data Analysis, Advanced SQL, Statistics, Machine Learning, Power BI and DAX, ETL and Data Warehousing.

Bachelor of Information Technology

Faculty of Computers and Information

September 2019 – July 2023 | Grade: Excellent

- Coursework: Databases, Data Structures and Algorithms, Software Engineering, Web Development, Networking.

LANGUAGES

Arabic — Native | English — Professional Working Proficiency | Verified credentials: arafa-a.lovable.app/certifications